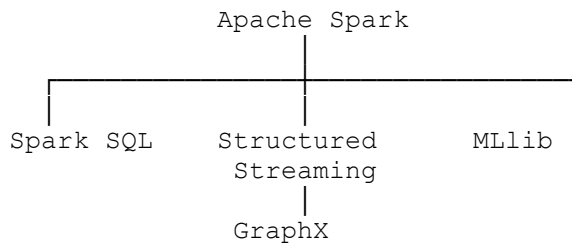


Apache Spark is a Unified Engine

Apache Spark is called a unified analytics engine because it provides **one engine** that can handle many different types of data processing without requiring separate frameworks.

Instead of using different tools for different tasks, Spark integrates them into a single platform.



Why is Spark called a unified engine?

Because all these components share the same:

- **Execution engine** (the same engine executes everything)
- **Cluster manager** (Standalone, YARN, Kubernetes, Mesos)
- **Programming APIs** (Python, Scala, Java, R)
- **Data abstractions** (RDDs, DataFrames, Datasets)
- **Optimization engine** (Catalyst + Tungsten for SQL/DataFrames)

This means you don't need different systems for SQL, streaming, and machine learning.

Example

Suppose you're building a fraud detection pipeline:

1. Read transaction data from Kafka using **Structured Streaming**.
2. Clean and transform the data using **DataFrames**.
3. Run SQL queries with **Spark SQL**.
4. Predict fraud using an **MLlib** model.
5. Save the results to a data lake.

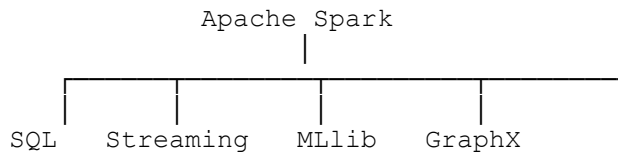
All of this runs inside **one Spark application**, using the **same execution engine**.

Before Spark (different engines)

SQL Processing	→ Hive
Streaming	→ Storm
Machine Learning	→ Mahout
Graph Processing	→ Giraph

Each tool had its own APIs, execution engine, and deployment model.

With Spark (one unified engine)



Everything runs on the same engine, making development, maintenance, and performance much simpler.

Interview answer

Apache Spark is a unified analytics engine because it provides a single execution engine that supports batch processing, SQL queries, real-time stream processing, machine learning, and graph processing through one programming model and one cluster management system.